2.1 MATRIX OPERATIONS

Three are two ways to denote an m x n matrix A:

- In terms of the columns of A $\quad A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \ldots \ \mathbf{a}_n]$
- In terms of the entries of A $\quad A = \begin{bmatrix} a_{11} & \ldots & a_{1j} & \ldots & a_{1n} \\ \vdots & \ldots & \vdots & \ldots & \vdots \\ a_{i1} & \ldots & a_{ij} & \ldots & a_{in} \\ \vdots & \ldots & \vdots & \ldots & \vdots \\ a_{m1} & \ldots & a_{mj} & \ldots & a_{mn} \end{bmatrix}$
- a_{ij} is the ith row and the jth column
- Diagonal entries are the entries on the main diagonal a_{11} , a_{22}, a_{33}. . .

The zero matrix is a matrix whose entries are all zeros $\quad \mathbf{0} = \begin{bmatrix} 0_{11} & \ldots & 0_{1j} & \ldots & 0_{1n} \\ \vdots & \ldots & \vdots & \ldots & \vdots \\ 0_{i1} & \ldots & 0_{ij} & \ldots & 0_{in} \\ \vdots & \ldots & \vdots & \ldots & \vdots \\ 0_{m1} & \ldots & 0_{mj} & \ldots & 0_{mn} \end{bmatrix}$

Definition: A diagonal matrix is a square matrix whose non-diagonal entries are zero.

Equal matrices has the same size and corresponding entries are equal.

In order to sum two matrices, the matrices must be the same size.

The scalar multiple, rA, is a matrix in which each entry is multiplied by the scalar r.

Ex 1. Given $\quad A = \begin{bmatrix} 2 & 0 & -1 \\ 4 & -5 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 7 & -5 & 1 \\ 1 & -4 & -3 \end{bmatrix}$

Find $\quad$ a) $-2A$ $\quad$ b) $B - 2A$

Theorem 1: Let A, B, and C be matrices of the same size and let r and s be scalars.

a. $A + B = B + A$
b. $(A + B) + C = A + (B + C)$
c. $A + \mathbf{0} = A$
d. $r(A + B) = rA + rB$
e. $(r + s)A = rA + sA$
f. $r(sA) = (rs)A$

MATRIX MULTIPLICATION

Definition: If A is an m x n matrix, and if B is a n x p matrix with columns $\mathbf{b}_1 \ldots \mathbf{b}_p$, then the product AB is the m x p matrix whose columns are $A\mathbf{b}_1 \ldots A\mathbf{b}_p$.

Ex 2: If A is a 4 x 3 matrix and B is a 3 x 2 matrix, then what are the sizes of AB and BA?

Suppose $A = \begin{bmatrix} 2 & 3 & 6 \\ -1 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -3 \\ 0 & 1 \\ 4 & -7 \end{bmatrix}$,

To compute AB, first find the entry in the first row-first column: multiply the entries in the 1st row of the 1st matrix times the corresponding entries in the 1st column of the 2nd matrix.

$$\begin{bmatrix} \mathbf{2} & \mathbf{3} & \mathbf{6} \\ -1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 2 & -3 \\ 0 & 1 \\ 4 & -7 \end{bmatrix} = \begin{bmatrix} 28 & \mathbf{-45} \\ \blacksquare & \blacksquare \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 6 \\ \mathbf{-1} & \mathbf{0} & \mathbf{1} \end{bmatrix}\begin{bmatrix} \mathbf{2} & -3 \\ \mathbf{0} & 1 \\ \mathbf{4} & -7 \end{bmatrix} = \begin{bmatrix} 28 & -45 \\ \mathbf{2} & \blacksquare \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 6 \\ \mathbf{-1} & \mathbf{0} & \mathbf{1} \end{bmatrix}\begin{bmatrix} 2 & \mathbf{-3} \\ 0 & \mathbf{1} \\ 4 & \mathbf{-7} \end{bmatrix} = \begin{bmatrix} 28 & -45 \\ 2 & \mathbf{-4} \end{bmatrix}$$

So $AB = \begin{bmatrix} 28 & -45 \\ 2 & -4 \end{bmatrix}$.

Ex 3: Given $C = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$ $D = \begin{bmatrix} 3 & 5 \\ -1 & 4 \end{bmatrix}$ $E = \begin{bmatrix} -5 \\ 3 \end{bmatrix}$

Find a. *CD* b. *DC* c. *DE* d. *ED*

PROPERTIES OF MATRIX MULTIPLICATION

Theorem 2: Let *A* be an *m* x *n* matrix and let *B* and *C* have sizes for which the indicated sums and products are defined.

(a) $A(BC) = (AB)C$
(associative law for multiplication)

(b) $A(B + C) = AB + AC$
(left distributive law)

(c) $(B + C)A = BA + CA$
(right distributive law)

(d) $r(AB) = (rA)B = A(rB)$ for any scalar r

(e) $I_m A = A = AI_n$
(identity for matrix multiplication)

WARNINGS

1. In general, $AB \neq BA$.
2. The cancellation laws do not hold for matrix multiplication. If $AB = AC$, then it is not true in general that $B = C$.
3. If a product AB is the zero matrix, you cannot conclude in general that either $A = 0$ or $B = 0$.

Ex 4: If $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, find a nonzero matrix B such that $AB = 0$.

POWERS OF A MATRIX

$$A^k = \underbrace{A \cdot A \cdots A}_{k \text{ times}}$$

For example,

$$\begin{bmatrix}1 & 0\\ 3 & 2\end{bmatrix}^3 = \begin{bmatrix}1 & 0\\ 3 & 2\end{bmatrix}\begin{bmatrix}1 & 0\\ 3 & 2\end{bmatrix}\begin{bmatrix}1 & 0\\ 3 & 2\end{bmatrix} =$$

$$\begin{bmatrix}__ & __\\ __ & __\end{bmatrix}\begin{bmatrix}1 & 0\\ 3 & 2\end{bmatrix} = \begin{bmatrix}1 & 0\\ 21 & 8\end{bmatrix}$$

The Transpose of a Matrix

Given an m x n matrix A, the transpose of A is the n x m matrix denoted by A^T, whose columns are formed from the corresponding rows of A.

For example, if $A = \begin{bmatrix}1 & 2 & 3 & 4\\ 5 & 6 & 7 & 8\\ 9 & 0 & -1 & -2\end{bmatrix}$, then $A^T = \begin{bmatrix}1 & 5 & 9\\ 2 & 6 & 0\\ 3 & 7 & -1\\ 4 & 8 & -2\end{bmatrix}$

Theorem 3: Let A and B denote matrices whose sizes are appropriate for the following sums and products.

(a) $(A^T)^T = A$

(b) $(A + B)^T = A^T + B^T$

(c) For any scalar r, $(rA)^T = rA^T$.

(d) $(AB)^T = B^TA^T$

**The transpose of a product of matrices equals the product of their transposes in the reverse order.

Ex 5: Given $A=\begin{bmatrix}1 & 2 & 0\\ 3 & 0 & 1\end{bmatrix}$ and $B=\begin{bmatrix}1 & 2\\ 0 & 1\\ -2 & 4\end{bmatrix}$, find each of the following, if possible.

a. $(AB)^T$

b. $(A^TB^T)^2$

c. B^TA^T

Ex 6: Prove $(ABC)^T = C^TB^TA^T$

(Hint: Use part d of Theorem 3)

$(ABC)^T =$

2.2 THE INVERSE OF A MATRIX

The inverse of a real number a is denoted by a^{-1}. For example, $7^{-1}=\frac{1}{7}$ and $7\cdot 7^{-1}=7^{-1}\cdot 7=1$

An $n\times n$ matrix A is said to be invertible if there is an $n\times n$ matrix C satisfying $CA = AC= I_n$ where I_n is the $n\times n$ identity matrix. We call C the inverse of A .

FACT: If A is invertible, then the inverse is unique.

Proof: Assume B and C are both inverses of A. Then, by definition $BA = I$ and $CA = I$

$$\begin{aligned} B &= BI \\ &= B(CA) \\ &= B(AC) \\ &= (BA)C \\ &= IC \\ &= C \end{aligned}$$

So the inverse is unique since any two inverses coincide.
The inverse of A is usually denoted by A^{-1}.

We have $AA^{-1}=A^{-1}A=I_n$

Not all $n\times n$ matrices are invertible. A matrix which is not invertible is sometimes called a singular matrix. An invertible matrix is called nonsingular matrix.

Ex 1: Given $A=\begin{bmatrix}-3 & -4\\ 1 & 1\end{bmatrix}$, find A^{-1}.

We know $AA^{-1} = I$, so $\begin{bmatrix}-3 & -4\\ 1 & 1\end{bmatrix}\begin{bmatrix}a & b\\ c & d\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$.

Set up a system of equations and solve it.

Theorem 4: Let $A=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and $A^{-1}=\frac{1}{ad-bc}\begin{bmatrix}d & -b\\ -c & a\end{bmatrix}$.

If $ad - bc = 0$, then A is not invertible.

Proof: $If\ A=\begin{bmatrix}a & b\\ c & d\end{bmatrix},\ then\ AA^{-1}=I\ if\ ad\neq bc$

$Let\ A^{-1}=\begin{bmatrix}x_1 & x_2\\ x_3 & x_4\end{bmatrix},\ then\ AA^{-1}=\begin{bmatrix}ax_1+bx_3 & ax_2+bx_4\\ cx_1+dx_3 & cx_2+dx_4\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

Complete the proof.

The determinant of A is $ad - bc$.

$$\det A = ad - bc$$

Matrix A is invertible if and only if $\det A\neq 0$.

Ex 2: Find the inverse of $\begin{bmatrix}8 & 6\\ 5 & 4\end{bmatrix}$.

Theorem 5: If A is an invertible n x n matrix, then for each b in R^n, the equation $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$.

Proof: Given $A\mathbf{x} = \mathbf{b}$

$A^{-1}A\mathbf{x} =$ _____

____$\mathbf{x} =$ _____

$\mathbf{x} = A^{-1}\mathbf{b}$

Ex 3: Use the inverse found in Ex 2. to solve the system $8x_1 + 6x_2 = 2$
$5x_1 + 4x_2 = -1$

Theorem 6:

(a) If A is an invertible matrix, then A^{-1} is invertible and $(A^{-1})^{-1} = A$.
(b) If A and B are n x n invertible matrices, then so is AB, and the inverse of AB is the product of the inverses of A and B in the reverse order: $(AB)^{-1} = B^{-1}A^{-1}$
(c) If A is an invertible matrix, then so is A^T, and the inverse of A^T is the transpose of A^{-1}: $(A^T)^{-1} = (A^{-1})^T$

Ex 4: Given $A = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}$, compute each of the following.

a. $A^{-1} + A\,A^{-1}$
b. $BC - (C^{-1}B^{-1})^{-1}$
c. $(BC)^T$

ELEMENTARY MATRICES

An elementary matrix is one that is obtained by performing a single elementary row operation on an identity matrix.

Examples: $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Elementary matrices are invertible because row operations are reversible. To determine the inverse of an elementary matrix E, determine the elementary row operation needed to transform E back into I and apply this operation to I to find the inverse.

If an elementary row operation is performed on an m x m matrix A, the resulting matrix can be written as EA, where the m x m matrix E is created by performing the same row operation on I_m.

For example, the matrix $\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}$ is an elementary matrix that was found by multiplying the second row by -3.
So, its inverse is found by applying the inverse row operation; i.e., multiply the second row of I_2 by -1/3. So, the inverse is $\begin{bmatrix} 1 & 0 \\ 0 & -1/3 \end{bmatrix}$.

You can check by multiplying $\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1/3 \end{bmatrix} = I_2$

Ex 5: Find the inverses of the previously given elementary matrices:

$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Each elementary matrix E is invertible.

The inverse of E is the elementary matrix of the same type that transforms E back into I.

Theorem 7: An n x n matrix A is invertible if and only if A is row equivalent to I_n, and in this case, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1}.

An Algorithm for Finding A^{-1}

Row reduce the augmented matrix $[A\ I]$. If A is row equivalent to I, then $[A\ I]$ is row equivalent to $[I\ A^{-1}]$. Otherwise, A does not have an inverse.

Ex 6: Find the inverse of $A = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}$ Hint: Start with $[AI] = \begin{bmatrix} 1 & 4 & 1 & 0 \\ -1 & -3 & 0 & 1 \end{bmatrix}$ and translate it to $[I\ A^{-1}]$.

Ex 7: Find the inverse of $A = \begin{bmatrix} 2 & 0 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ using the same algorithm.

Order of multiplication is important!

Ex 8: Suppose A,B,C, and D are invertible $n \times n$ matrices and $A=B(D-I_n)C$.

Solve for D in terms of A,B,C and I.

$A=B(D-I_n)C$
$B^{-1}A=$ continue……

2.3 Characterizations of Invertible Matrices

Theorem 8 (The Invertible Matrix Theorem) **Note:** Assume all matrices on this WS are square. Let A be a square n x n matrix. Then the following statements are equivalent (i.e., for a given A, they are either all true or all false).

a. A is an invertible matrix.
b. A is row equivalent to I_n.
c. A has n pivot positions.
d. The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
e. The columns of A form a linearly independent set.
f. The linear transformation $\mathbf{x} \rightarrow A\mathbf{x}$ is one-to-one.
g. The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbf{R}_n$.
h. The columns of A span $\mathbf{R}_n$.
i. The linear transformation $\mathbf{x}=A\mathbf{x}$ maps $\mathbf{R}_n$ onto $\mathbf{R}_n$.
j. There is an n x n matrix C such that $CA = I_n$.
k. There is an n x n matrix D such that $AD = I_n$.
l. A^T is an invertible matrix.

Determine which of the matrices below are invertible. Use as few calculations as possible. Justify your answers.

1. $\begin{bmatrix} 5 & 0 & 0 \\ -3 & -7 & 0 \\ 8 & 5 & -1 \end{bmatrix}$

2. $\begin{bmatrix} -7 & 0 & 4 \\ 3 & 0 & -1 \\ 2 & 0 & 9 \end{bmatrix}$

3. $\begin{bmatrix} 1 & -5 & -4 \\ 0 & 3 & 4 \\ -3 & 6 & 0 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 3 & 7 & 4 \\ 0 & 5 & 9 & 6 \\ 0 & 0 & 2 & 8 \\ 0 & 0 & 0 & 10 \end{bmatrix}$

5. $\begin{bmatrix} 5 & 3 & 1 & 7 & 9 \\ 6 & 4 & 2 & 8 & -8 \\ 7 & 5 & 3 & 10 & 9 \\ 9 & 6 & 4 & -9 & -5 \\ 8 & 5 & 2 & 11 & 4 \end{bmatrix}$ (calculator OK)

True/False **Note:** Assume all matrices on this WS are square.

6. If there is an *n x n* matrix *D* such that *AD = I*, then there is also an *n x n* matrix *C* such that *CA = I.*

7. If the columns of *A* are linearly independent, then the columns of *A* span R^n.

8. If the equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in R^n, then the solution is unique for each $\mathbf{b}$.

9. If the linear transformation maps R^n onto R^n then *A* has *n* pivot positions.

10. If there is a $\mathbf{b}$ in R^n such that the equation $A\mathbf{x} = \mathbf{b}$ is inconsistent, then the transformation $\mathbf{x}=A\mathbf{x}$ is not one-to-one.

Questions

11. If C is 6 x 6 and the equation $C\mathbf{x} = \mathbf{v}$ is consistent for every $\mathbf{v}$ in R^6, is it possible that for some $\mathbf{v}$, the equation $C\mathbf{x} = \mathbf{v}$ has more than one solution? Explain.

12. If the columns of a 7 x 7 matrix D are linearly independent, what can you say about the solutions of $D\mathbf{x} = \mathbf{b}$? Explain.

13. If $n \times n$ matrices E and F have the property that $EF = I$, then E and F commute. Why?

14. If the equation $G\mathbf{x} = \mathbf{y}$ has more than one solution for some $\mathbf{y}$ in R^n, can the columns of G span R^n? Explain.

15. If the equation $H\mathbf{x} = \mathbf{c}$ is inconsistent for some $\mathbf{c}$ in R^n, what can you say about the equation $H\mathbf{x} = \mathbf{0}$? Explain.

2.4 Partitioned Matrices

A matrix can be subdivided or partitioned into smaller matrices by inserting horizontal and vertical rules between selected rows and columns.

For example, below are three possible partitions of a general 3 x 4 matrix A.

The first is a partition of A into four submatrices A_{11}, A_{12}, A_{21}, A_{22}; the second is a partition of A into its row matrices $\mathbf{r}_1$, $\mathbf{r}_2$, and $\mathbf{r}_3$; the third is a partition of A into its column matrices $\mathbf{c}_1$, $\mathbf{c}_2$, $\mathbf{c}_3$ and $\mathbf{c}_4$.

$$A = \left[\begin{array}{cc|cc} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ \hline a_{31} & a_{32} & a_{33} & a_{34} \end{array}\right] = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

$$A = \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \hline a_{21} & a_{22} & a_{23} & a_{24} \\ \hline a_{31} & a_{32} & a_{33} & a_{34} \end{array}\right] = \begin{bmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \mathbf{r}_3 \end{bmatrix}$$

$$A = \left[\begin{array}{c|c|c|c} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{array}\right] = [\mathbf{c}_1 \quad \mathbf{c}_2 \quad \mathbf{c}_3 \quad \mathbf{c}_4]$$

Partitioned matrices appear in most modern applications of linear algebra.

Ex 1: $\left[\begin{array}{ccc:c:cc} 1 & 2 & 3 & 4 & 5 & 6 \\ 7 & 8 & -1 & 0 & 2 & 4 \\ \hdashline -9 & 5 & 8 & 3 & 1 & 7 \end{array}\right]$

a. Name A_{13}.

b. What are the dimensions of A_{22}?

Addition and Scalar Multiplication

If matrices A and B are the same size and are partitioned in exactly the same way, then it is natural to make the same partition of the ordinary matrix sum $A + B$

Each block of $A + B$ is the sum of the corresponding blocks of A and B.

Multiplication of a partitioned matrix by a scalar is also computed block by block.

Multiplication of Partitioned Matrices

Partitioned matrices can be multiplied by the usual row-column rule as if the block entries were scalars, provided that for a product AB, the column partition of A matches the row partition of B.

For example, if A and B are partitioned into submatrices $A = \left[\begin{array}{c:c} A_{11} & A_{12} \\ \hdashline A_{21} & A_{22} \end{array}\right]$ and $B = \left[\begin{array}{c:c} B_{11} & B_{12} \\ \hdashline B_{21} & B_{22} \end{array}\right]$ then AB can be expressed as

$$AB = \left[\begin{array}{c:c} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ \hdashline A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{array}\right]$$

In our A and B, the partitions are conformable for block multiplication

Ex 2: Multiply

$$\left[\begin{array}{ccc:cc} 1 & 2 & 3 & -1 & 0 \\ 2 & 4 & 1 & -2 & 3 \\ \hdashline 0 & 5 & 2 & 1 & 0 \end{array}\right] \left[\begin{array}{cc} -1 & -2 \\ 3 & 4 \\ 0 & 6 \\ \hdashline 1 & 0 \\ 2 & -3 \end{array}\right]$$

Row-column rule for multiplication of block matrices provides the most general way to regard the product of two matrices.

Theorem 10: Column-Row Expansion of AB

If A is m x n and B is n x p, then $AB = \begin{bmatrix} col_1(A) & col_2(A) & \cdots & col_n(A) \end{bmatrix} \begin{bmatrix} row_1(B) \\ row_2(B) \\ \vdots \\ row_n(B) \end{bmatrix} = = col_1(A)row_1(B) + \cdots + col_n(A)row_n(B)$

In Ex 3 and 4, assume that the matrices are partitioned conformably for block multiplication, O represents a matrix of zeros in every entry, and I is the identity. Compute the products.

Ex 3: $\begin{bmatrix} E & O \\ O & F \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix}$

Ex 4: $\begin{bmatrix} I & O \\ -X & I \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix}$

In Ex 5 and 6, find formulas for X, Y, and Z in terms of A, B, and C. (Hint: Compute the product on the left, and set it equal to the right). Assume each submatrix is square.

Ex 5:

$$\begin{bmatrix} X & O \\ Y & Z \end{bmatrix}\begin{bmatrix} A & O \\ B & C \end{bmatrix}=\begin{bmatrix} I & O \\ O & I \end{bmatrix}$$

Ex 6: $\begin{bmatrix} A & B \\ O & I \end{bmatrix}\begin{bmatrix} X & Y & Z \\ O & O & I \end{bmatrix}=\begin{bmatrix} I & O & O \\ O & O & I \end{bmatrix}$

Inverses of Partitioned Matrices

Block upper triangular $A=\begin{bmatrix} A_{11} & A_{12} \\ O & A_{22} \end{bmatrix}$

Assume A_{11} is p x p, $A_{22}$is q x q, and A is invertible. Find a formula for A^{-1}

Start with the equation $\begin{bmatrix} A_{11} & A_{12} \\ O & A_{22} \end{bmatrix}\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}=\begin{bmatrix} I_p & O \\ O & I_q \end{bmatrix}$

A block diagonal matrix is a partitioned matrix with zero blocks off the main diagonal (of blocks). Such a matrix is invertible if and only if each block on the diagonal is invertible.

The partitioned matrix $\begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix}$ is a block diagonal matrix.

2.5 MATRIX FACTORIZATION

A **factorization** of a matrix A is an equation that expresses A as a product of two or more matrices.

- Matrix multiplication -- synthesis of data
- Matrix factorization – analysis of data

In the language of computer science the expression of A as a product amounts to a **preprocessing** of the data in A, organizing that data into two or more parts whose structures are more useful in some way, perhaps more accessible for computation.

This method is well suited for computers and is the basis for many practical computer applications. Programs such as MatLab, Mathematica and Maple utilize LU factorizations.

The LU Factorization

We will discuss a factorization approach based on factoring the coefficient matrix into a product of lower and upper triangular matrices.

In particular, L is an m x n lower triangular matrix with 1's on the diagonal. U is an m x n echelon form of A.

$$A = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} \square & * & * & * & * \\ 0 & \square & * & * & * \\ 0 & 0 & 0 & \square & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_{U}$$

How do we perform an LU factorization to start this whole process?

Step 1: Reduce A to an echelon form U by a sequence of operations without row interchanges.

Step 2: Place entries in L such that the same sequence of row operations reduced L to I

Example: We will reduce $\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix}$ to its LU factorization.

$$\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} \text{ ½ R1}$$

$$\begin{bmatrix} 1 & 3 & 1 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 \\ -3 & -8 & 0 \\ 0 & -3 & -2 \end{bmatrix} \text{ -4R1+R3}\rightarrow\text{R3}$$

$$\begin{bmatrix} 1 & 3 & 1 \\ -3 & -8 & 0 \\ 0 & -3 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & -3 & -2 \end{bmatrix} \text{3R1+R2}\rightarrow\text{R2}$$

$$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & -3 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 7 \end{bmatrix} \text{3R2+R3}\rightarrow\text{R3}$$

$$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 7 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \text{1/7 R3}\rightarrow\text{ R3}$$

Note the operations used and the order they were used:
½ R1
-4R1+R3→R3
3R1+R2→R2
3R2+R3→R3
1/7 R3→ R3

In order to find L, take I and perform the reversal of the operations

$$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}\rightarrow\begin{bmatrix}1&0&0\\0&1&0\\0&0&7\end{bmatrix}\ 7\ R3\rightarrow R3$$

$$\begin{bmatrix}1&0&0\\0&1&0\\0&0&7\end{bmatrix}\rightarrow\begin{bmatrix}1&0&0\\0&1&0\\0&-3&7\end{bmatrix}\ -3R2+R3\rightarrow R3$$

$$\begin{bmatrix}1&0&0\\0&1&0\\0&-3&7\end{bmatrix}\rightarrow\begin{bmatrix}1&0&0\\-3&1&0\\0&-3&7\end{bmatrix}\ -3R1+R2\rightarrow R2$$

$$\begin{bmatrix}1&0&0\\-3&1&0\\0&-3&7\end{bmatrix}\rightarrow\begin{bmatrix}1&0&0\\-3&1&0\\4&-3&7\end{bmatrix}\ 4R1+R3\rightarrow R3$$

$$\begin{bmatrix}1&0&0\\-3&1&0\\4&-3&7\end{bmatrix}\rightarrow\begin{bmatrix}2&0&0\\-3&1&0\\4&-3&7\end{bmatrix}\ 2\ R1$$

So, $L=\begin{bmatrix}2&0&0\\-3&1&0\\4&-3&7\end{bmatrix}$ Check by multiplying LU to get A

Ex 1: Find an LU decomposition of $\begin{bmatrix}6&-2&0\\9&-1&1\\3&7&5\end{bmatrix}$

Ex 2: Find an LU decomposition of $\begin{bmatrix}-5&3&4\\10&-8&-9\\15&1&2\end{bmatrix}$

If an n x n matrix A can be factored into a product of n x n matrices as $A = LU$ where L is a lower triangular and U is an upper triangular matrix, then the linear system $A\mathbf{x} = \mathbf{b}$ can be solved as follows:

Step 1: Rewrite the system $A\mathbf{x} = \mathbf{b}$ as $LU\mathbf{x} = \mathbf{b}$

Step 2: Define a new n x 1 matrix $\mathbf{y}$ by $U\mathbf{x} = \mathbf{y}$ so that $L\mathbf{y} = \mathbf{b}$. Solve for $\mathbf{y}$ with the augmented matrix $[L\mathbf{b}] = \mathbf{y}$

Step 3: Solve $U\mathbf{x} = \mathbf{y}$ by solving $[U\mathbf{y}] = \mathbf{x}$

Although this procedure replaces the problem of solving the single system $A\mathbf{x} = \mathbf{b}$ by the problem of solving the two systems $L\mathbf{y} = \mathbf{b}$ and $U\mathbf{x} = \mathbf{y}$, the latter systems are easy to solve because the coefficient matrices are triangle.

Example of solving a system by *LU* factorization :

Recall: $\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix}\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$

We will use this to solve the system

$$\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

Step 1: Rewrite the system $A\mathbf{x} = \mathbf{b}$ as $LU\mathbf{x} = \mathbf{b}$

$$\begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix}\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

L U $\mathbf{x}$ $\mathbf{b}$

Step 2: Define a new n x 1 matrix $\mathbf{y}$ by $U\mathbf{x} = \mathbf{y}$ so that $L\mathbf{y} = \mathbf{b}$. Solve for $\mathbf{y}$ with the augmented matrix $[L\mathbf{b}] = \mathbf{y}$

$$\begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

L $\mathbf{y}$ $\mathbf{b}$

So, $\begin{bmatrix} 2 & 0 & 0 & 2 \\ -3 & 1 & 0 & 2 \\ 4 & -3 & 7 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix}$

Thus, $\mathbf{y} = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$

Step 3: Solve $U\mathbf{x} = \mathbf{y}$ by solving $[U\mathbf{y}] = \mathbf{x}$

$$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$$

U $\mathbf{x}$ $\mathbf{y}$

So, $\begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$

Thus, $\mathbf{x} = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$

Ex 3: Solve the equation $A\mathbf{x} = \mathrm{b}$ by using the *LU* factorization given for *A*.

$$A = \begin{bmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{bmatrix},\ b = \begin{bmatrix} -7 \\ 5 \\ 2 \end{bmatrix} \quad \text{where } A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{bmatrix}\begin{bmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

Ex 4: Solve the equation $A\mathbf{x} = \mathrm{b}$ by using the LU factorization given for A.

$$A = \begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix},\ b = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix} \text{ where } A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}\begin{bmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

Ex 5: Solve $A\mathbf{x} = \mathbf{b}$ for $\mathbf{x}$ using LU factorization $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & -2 & 1 \\ -1 & 1 & 1 \end{bmatrix},\ b = \begin{bmatrix} 8 \\ 4 \\ -1 \end{bmatrix}$

Ex 6: Solve $A\mathbf{x} = \mathbf{b}$ for $\mathbf{x}$ using LU factorization $A = \begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix},\ b = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}$

2.7 APPLICATIONS TO COMPUTER GRAPHICS

A look at basic mathematics used to manipulate and display graphical images such as a wire-frame model of an airplane.

Math of computer graphics is closely connected with matrix multiplication but it does not correspond directly to matrix multiplication because translation is not a linear transformation

Homogeneous Coordinates

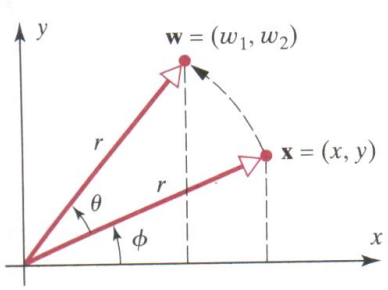
Each point (x, y) in R^2 can be identified with the point $(x, y, 1)$ on a plane in R^3 that lies one unit above the xy-plane.

(x, y) is said to have homogeneous coordinates $(x, y, 1)$.

Homogeneous coordinates for points are not added or multiplied by scalars, but they can be transformed via multiplication by a 3 x 3 matrix.

Ex: Translation $\begin{aligned}(x,y)&\mapsto(x+h,y+k)\\(x,y,1)&\mapsto(x+h,y+k,1)\end{aligned}$ is given by the matrix $\begin{bmatrix}1&0&h\\0&1&k\\0&0&1\end{bmatrix}\begin{bmatrix}x\\y\\1\end{bmatrix}=\begin{bmatrix}x+h\\y+k\\1\end{bmatrix}$

One of the most common and useful transformations is a rotation transformation or operator. An operator that rotates each vector in R^2 through a fixed angle θ is called a rotation operator on R^2.



To find equations relating a vector **x** to its transformation **w** = T(**x**), let ϕ be the angle from the positive x-axis to **x** and let r be the common length of **x** and **w**.

Consider the diagram at the right.

So, from basic trig, we get
$x = r\cos\theta$ and
$y = r\sin\theta$ and
$w_1 = r\cos(\theta + \phi)$ and
$w_2 = r\sin(\theta + \phi)$

Using trig identities yields
$w_1 = r\cos\theta\cos\phi - r\sin\theta\sin\phi$ and
$w_2 = r\sin\theta\sin\phi + r\cos\theta\sin\phi$

By substituting, we get
$w_1 = x\cos\theta - y\sin\theta$
$w_2 = x\sin\theta + y\cos\theta$

So, the rotation transformation is
$\begin{bmatrix}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{bmatrix}$

The corresponding homogeneous transformation is

Counter-clockwise rotation about the origin, angle φ $\begin{bmatrix}\cos\varphi&-\sin\varphi&0\\\sin\varphi&\cos\varphi&0\\0&0&1\end{bmatrix}$

The homogeneous transformation for a reflection through $y = x$ is $\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$

Note that the homogeneous transformations are created from the transformations learned in section 1.9.

A composite transformation – 2 or more basic transformations formed by multiplication of matrices.

$$\begin{bmatrix} Last \\ Trans. \end{bmatrix}\begin{bmatrix} 2nd \\ Trans. \end{bmatrix}\begin{bmatrix} 1st \\ Trans. \end{bmatrix}\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

In Ex 1 and 2, find a 3 x 3 matrix that produces the described composite 2D transformations, using homogeneous coordinates
Ex 1: Translate by (-2, 3) and then scale the *x*-coordinate by .8 and the *y*-coordinate by 1.2

Ex 2: Rotate points $30°$, and then reflect through the *x*-axis.

3D Computer Graphics

Newest and most exciting work in computer graphics is connected with molecular modeling.

Biologists can rotate and translate an experimental drug and attempt to attach it to a protein.
The ability to **visualize** potential chemical reactions is vital to modern drug and cancer research.

Current research in molecular modeling is focused on **virtual reality**, an environment in which a researcher can see and **feel** the drug molecule slide into the protein.
Another design for virtual reality involves a helmet and glove that detect head, hand and finger movements.

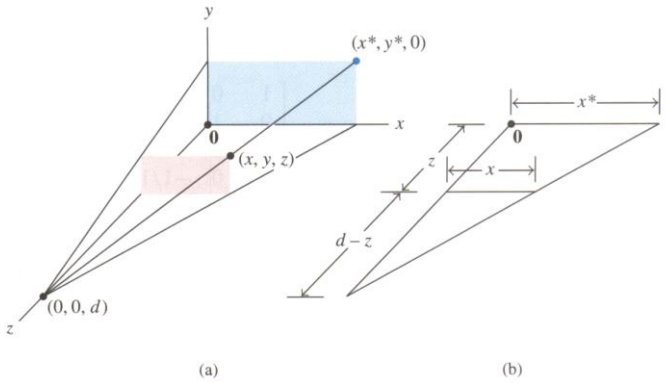
Homogeneous 3D Coordinates
$(x, y, z, 1)$ are homogeneous coordinates for the point (x, y, z) in R^3

In general, (X, Y, Z, H) are homogeneous coordinates for (x, y, z) if $H \neq 0$ and $x = \frac{X}{H}, y = \frac{Y}{H}, and\ z = \frac{Z}{H}$

Perspective Projections Extension: (Time permitting)

3D object is represented on the 2D computer screen by projecting the object onto a viewing plane.
For simplicity, let the *xy*-plane represent the computer screen, and imagine that the eye of a viewer is along the positive *x*-axis, at a point (0, 0, *d*).

A perspective projection maps each point (*x*, *y*, *z*) onto an image point $(x^*, y^*, 0)$ so that the two points and the eye position, called the center of projection, are on a line.



The triangle in the *xz*-plane in (a) is redrawn in (b) showing the lengths of line segments.

Similar triangles show that

$$\frac{x^*}{d} = \frac{x}{d-z} \quad \text{and} \quad x^* = \frac{dx}{d-z} = \frac{x}{1-\frac{z}{d}}. \text{ Similarly,}$$

$$y^* = \frac{y}{1-\frac{z}{d}}$$

Using homogeneous coordinates, perspective projection matrix P and we want (*x*, *y*, *z*, 1) to map into $\left(\frac{x}{1-\frac{z}{d}}, \frac{y}{1-\frac{z}{d}}, 0, 1\right)$

$$P = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{-1}{d} & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 0 \\ 1-\frac{z}{d} \end{bmatrix}$$

Ex 3: Let S be the triangle with vertices (9, 3, -5), (12, 8, 2), and (1.8, 2.7, 1). Find the image of S under the perspective projection with center of projection at (0, 0, 10).